BookHeaven Software Design Document (SDD)

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1. Introduction:

1.1/ Purpose:

- The purpose of this document is to outline the general structure and the inner mechanisms of **BookHeaven.**

1.2/Scope:

- This SDD encompasses the technical design for a Book Tracking and Management Website that allows users to manage the books they are currently reading, want to read, and have finished reading.

1.3/ Intended Audience:

- This document is intended for the developers.

1.4/ References:

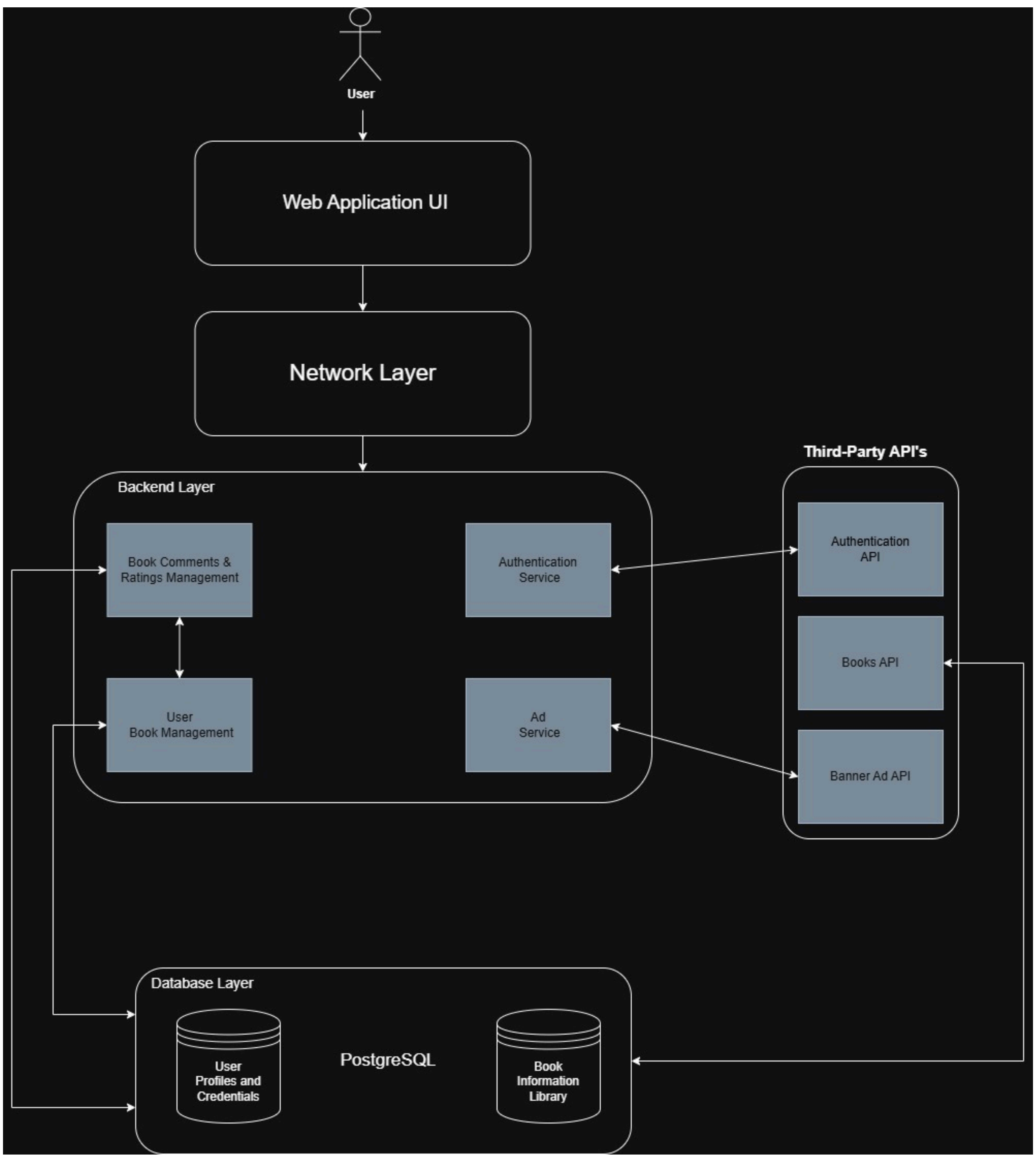
- **D1:** https://github.com/memoryVoid1/BookHeaven_Deliverables/blob/main/BookHeaven_D1_SDP.pdf
- **D2:** https://github.com/memoryVoid1/BookHeaven_Deliverables/blob/main/BookHeaven_D2_SRS.pdf

2. System Overview:

2.1/ Summary:

- **BookHeaven** is a standalone, browser-based book tracking web application that allows users to manage books they have read, are currently reading, and want to read. Key features include secure **user login**, **adding books to a collection**, **categorizing by reading status** (Read/Reading/To Be Read), and **adding notes/ratings**. The system will securely store user and book data using a database like **PostgreSQL** and will depend on a **Web Hosting Service**. There is potential for future integration with external book APIs (e.g., Google Books API).

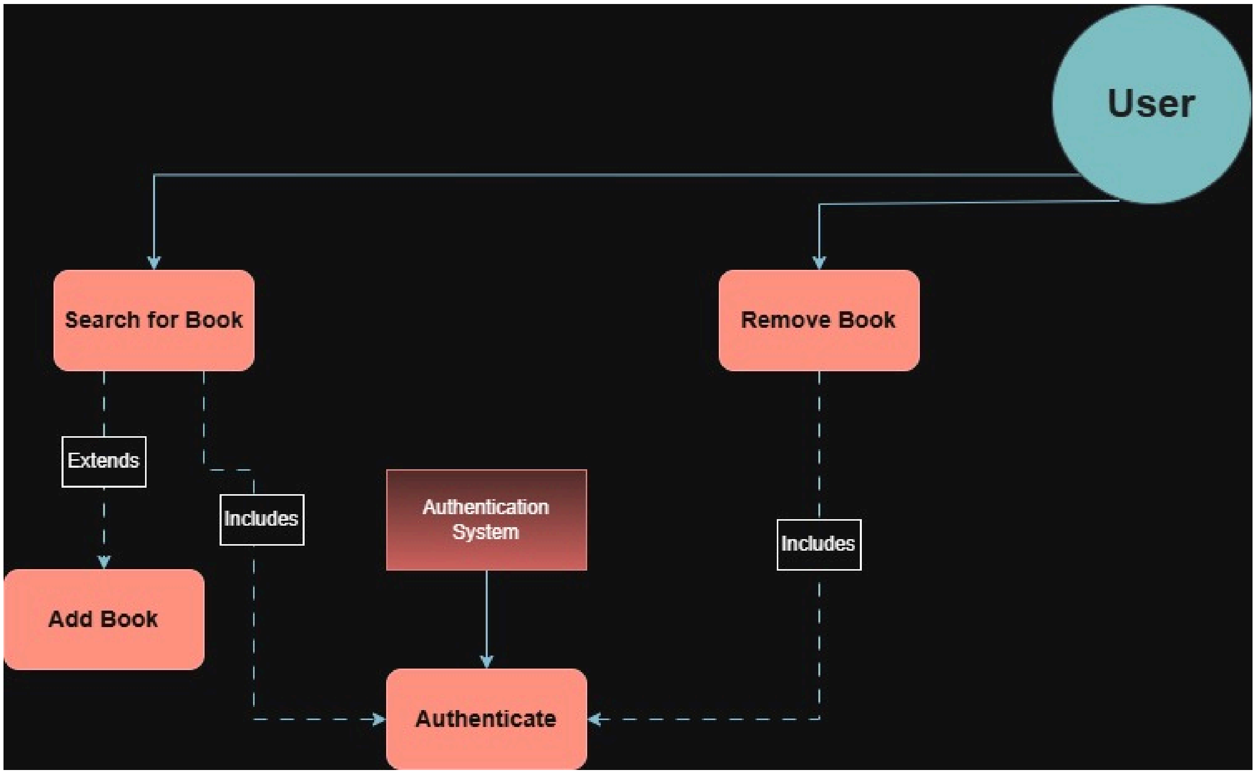
2.2/ System Diagram:



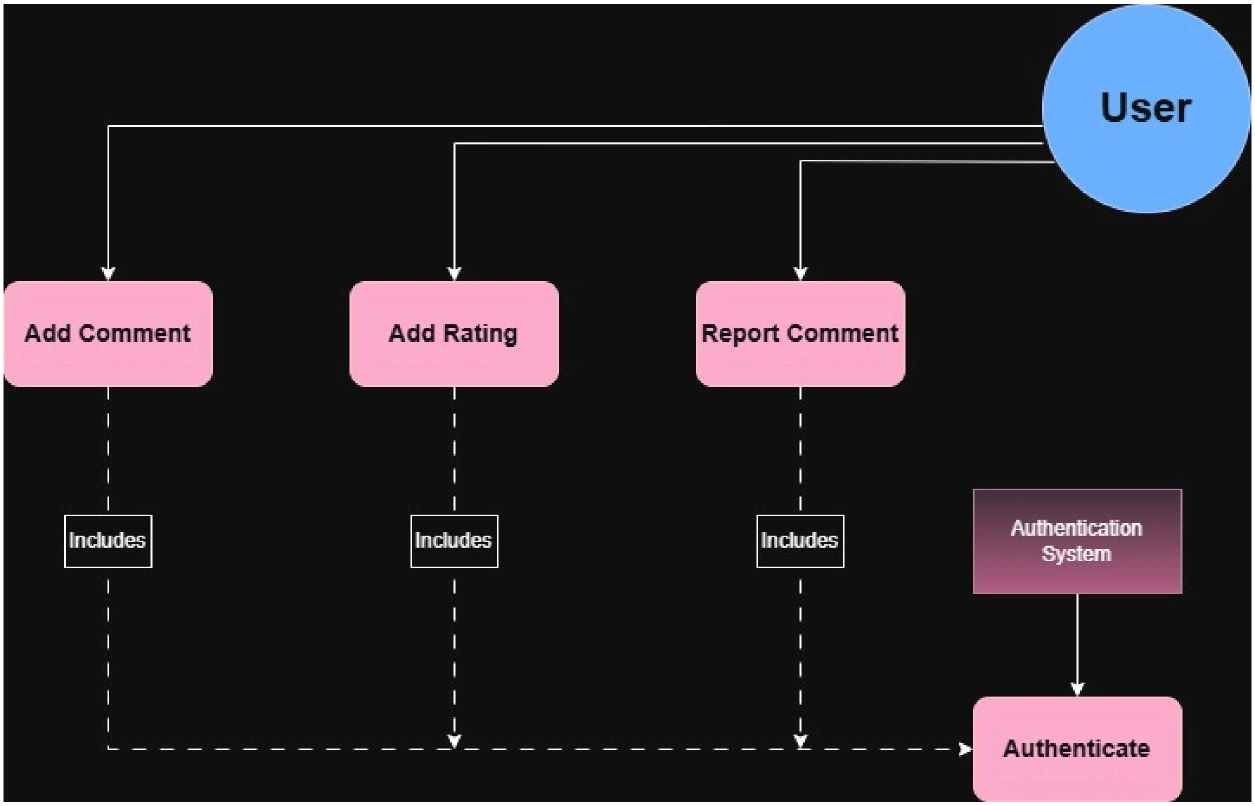
3. UML Diagrams:

3.1/ Use Case Diagrams:

Book Management:



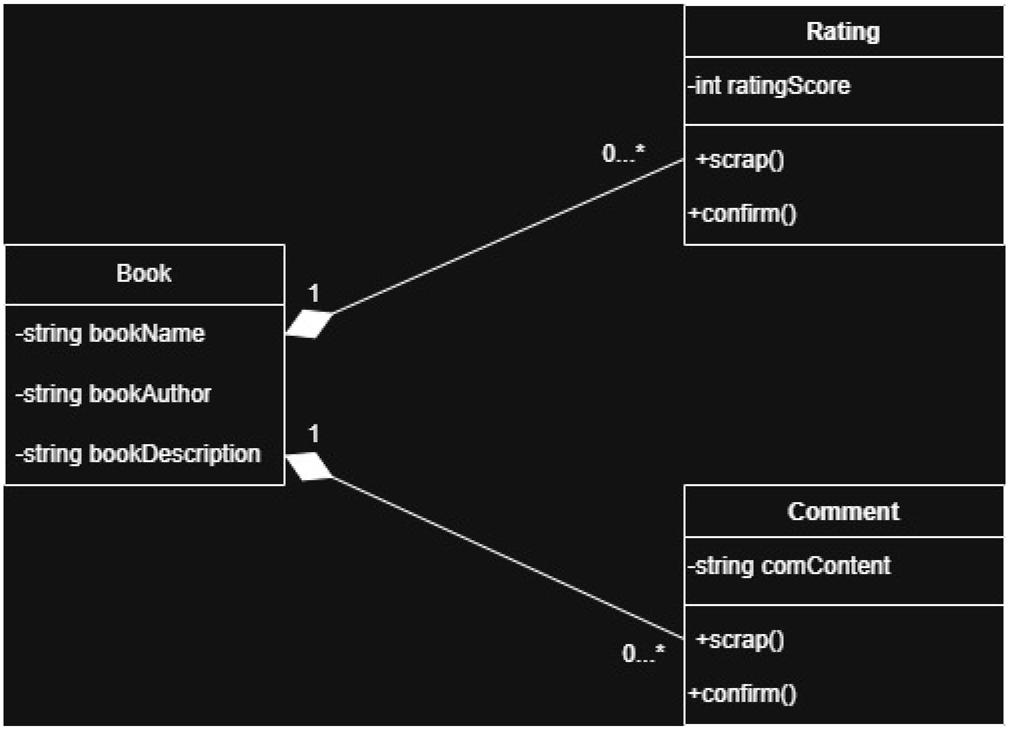
Comments and Rating:



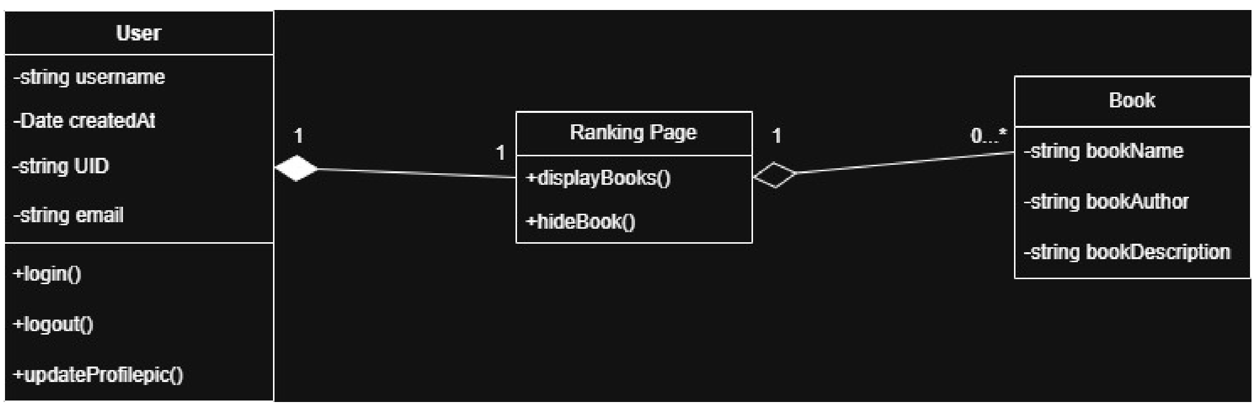
3.2/ Class Diagrams:

Book Management:

Comments and Rating:

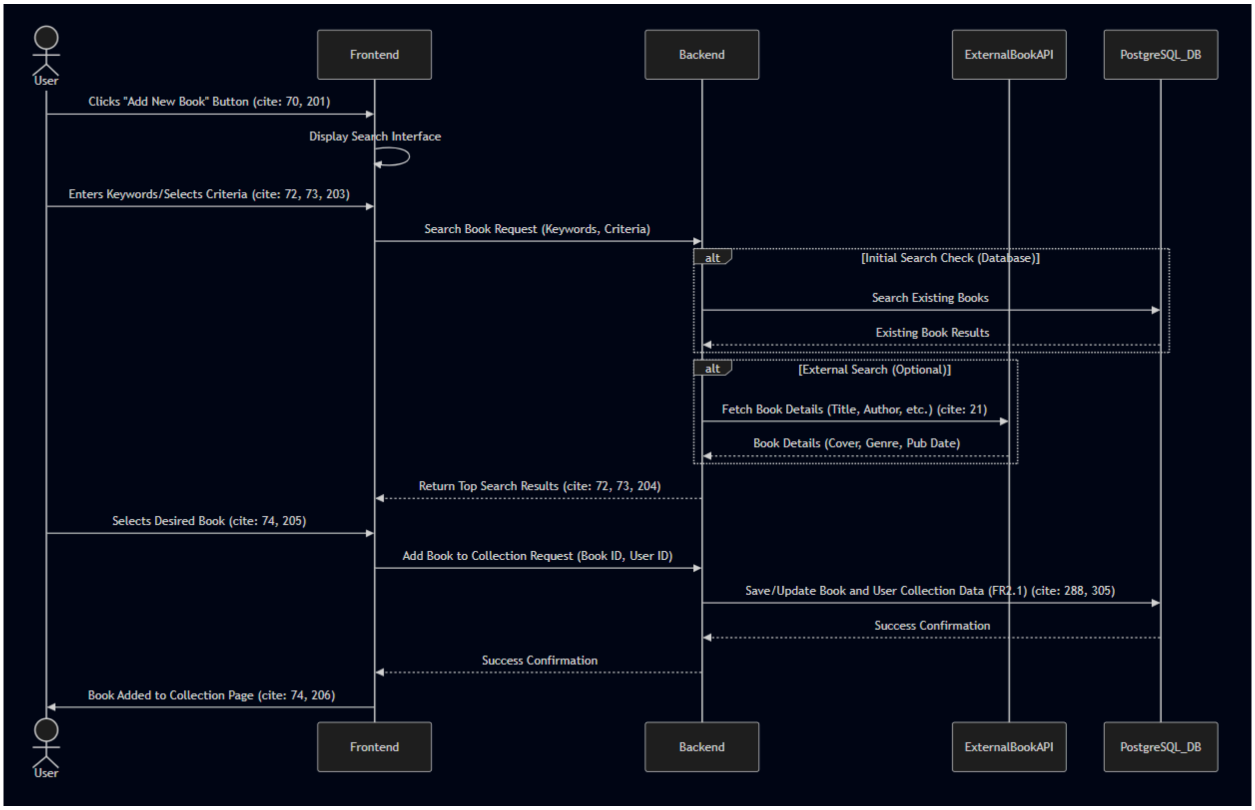


Ranking Page:

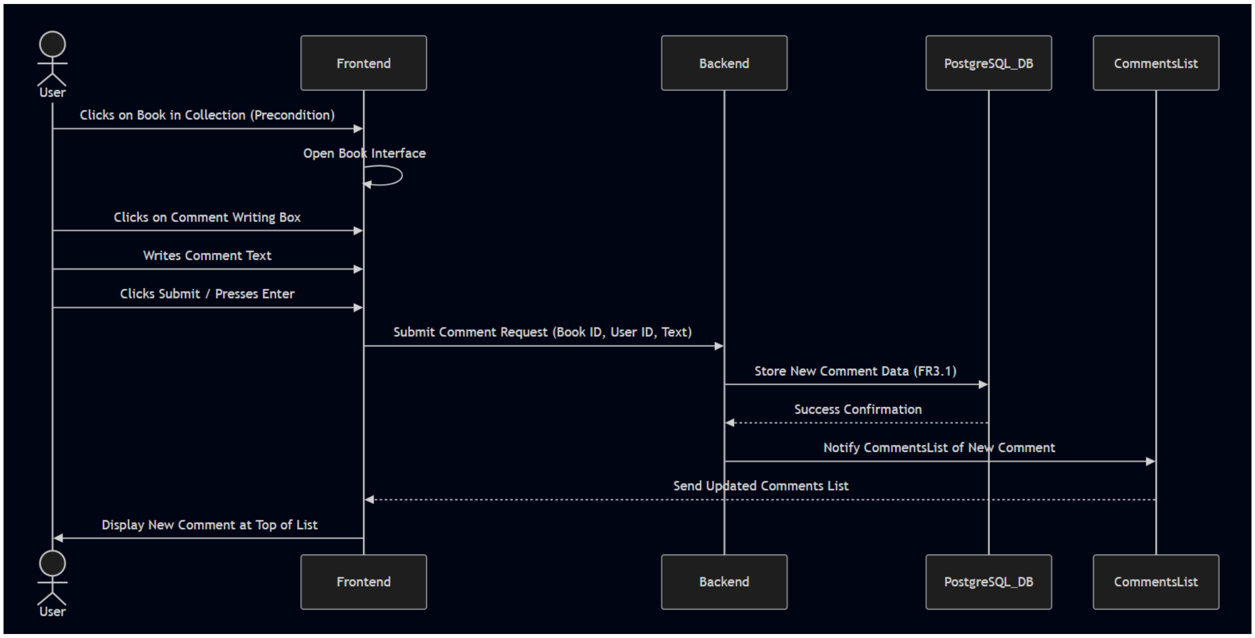


3.3/ Sequence Diagrams:

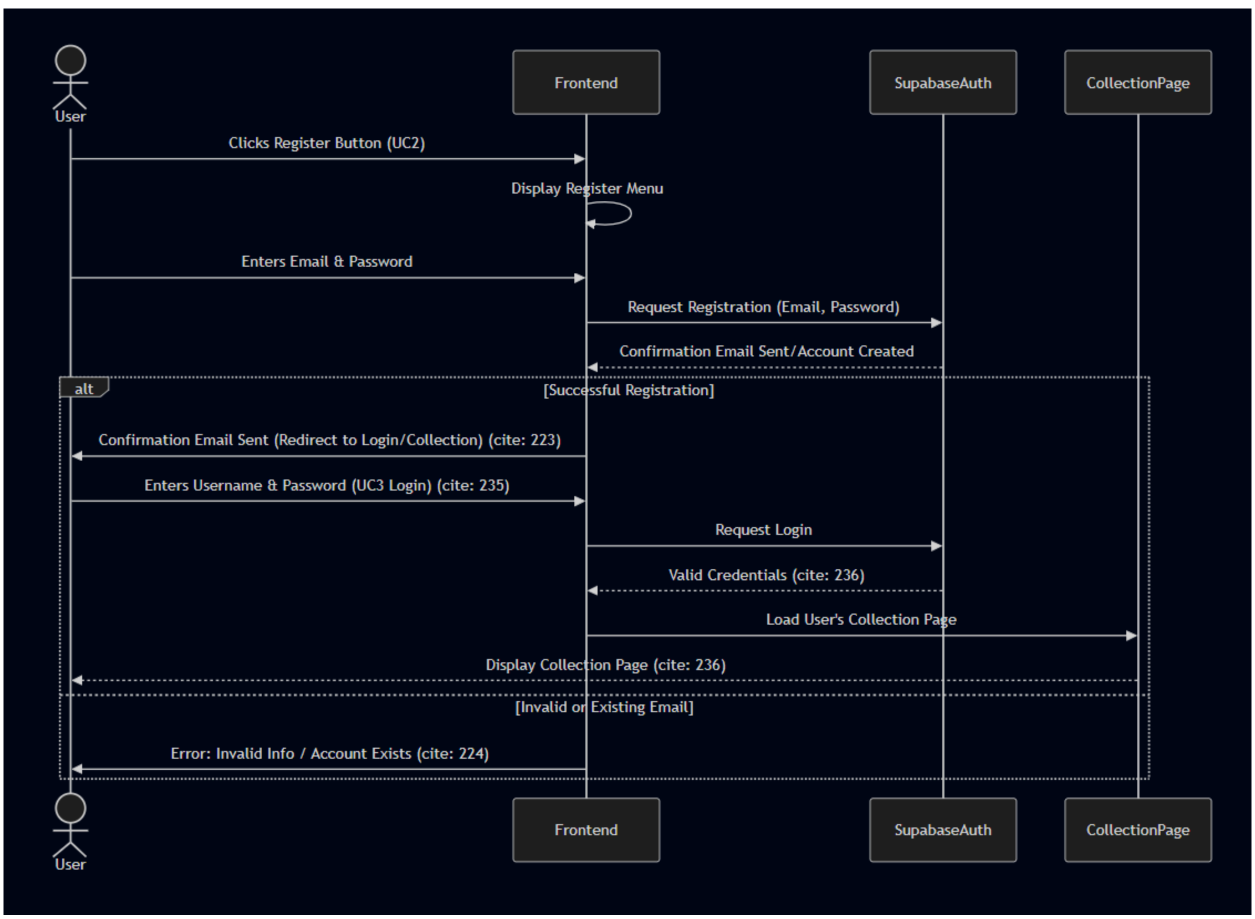
A/ Book Adding:



B/ Commenting:

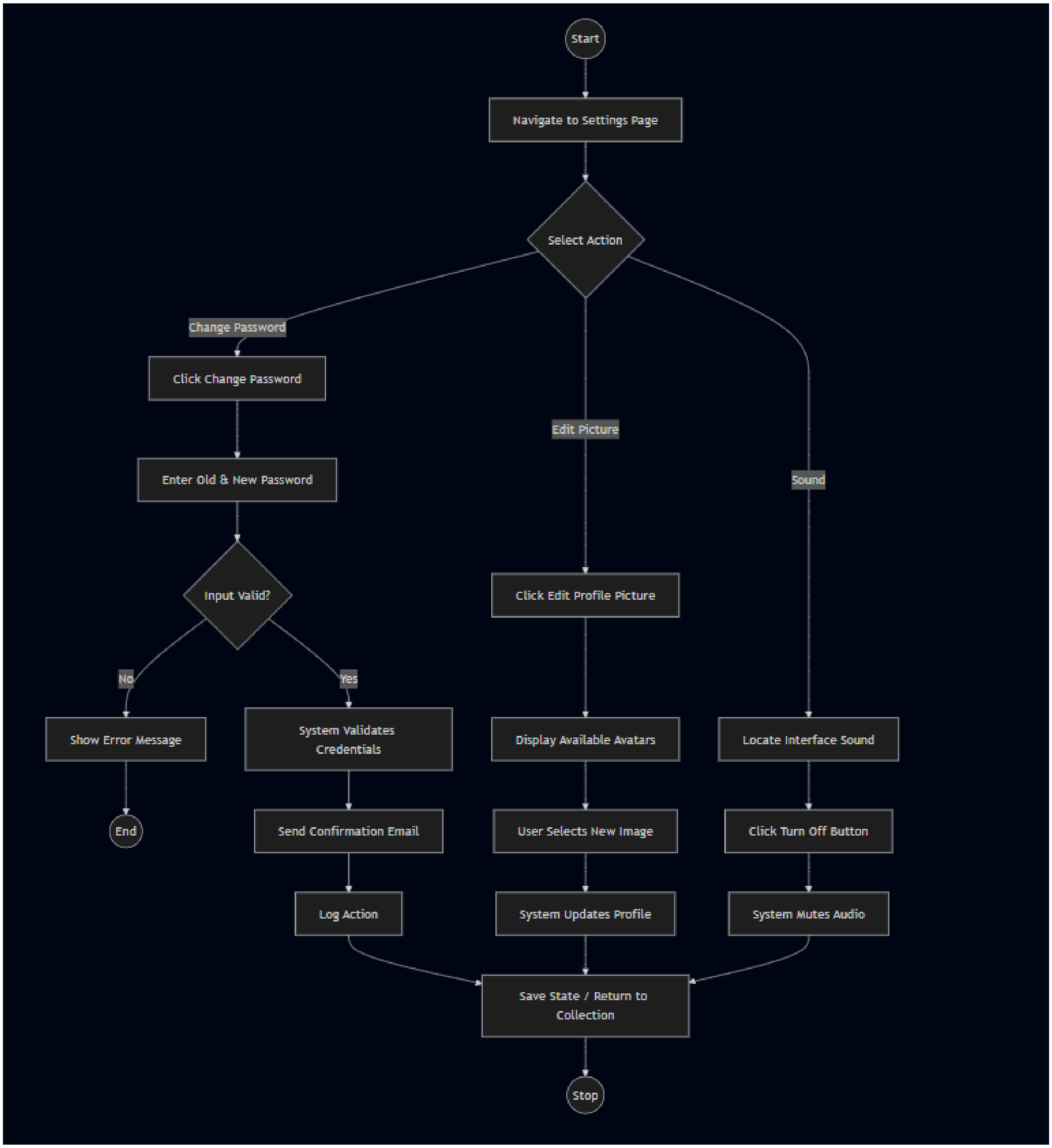


C/ Registering:

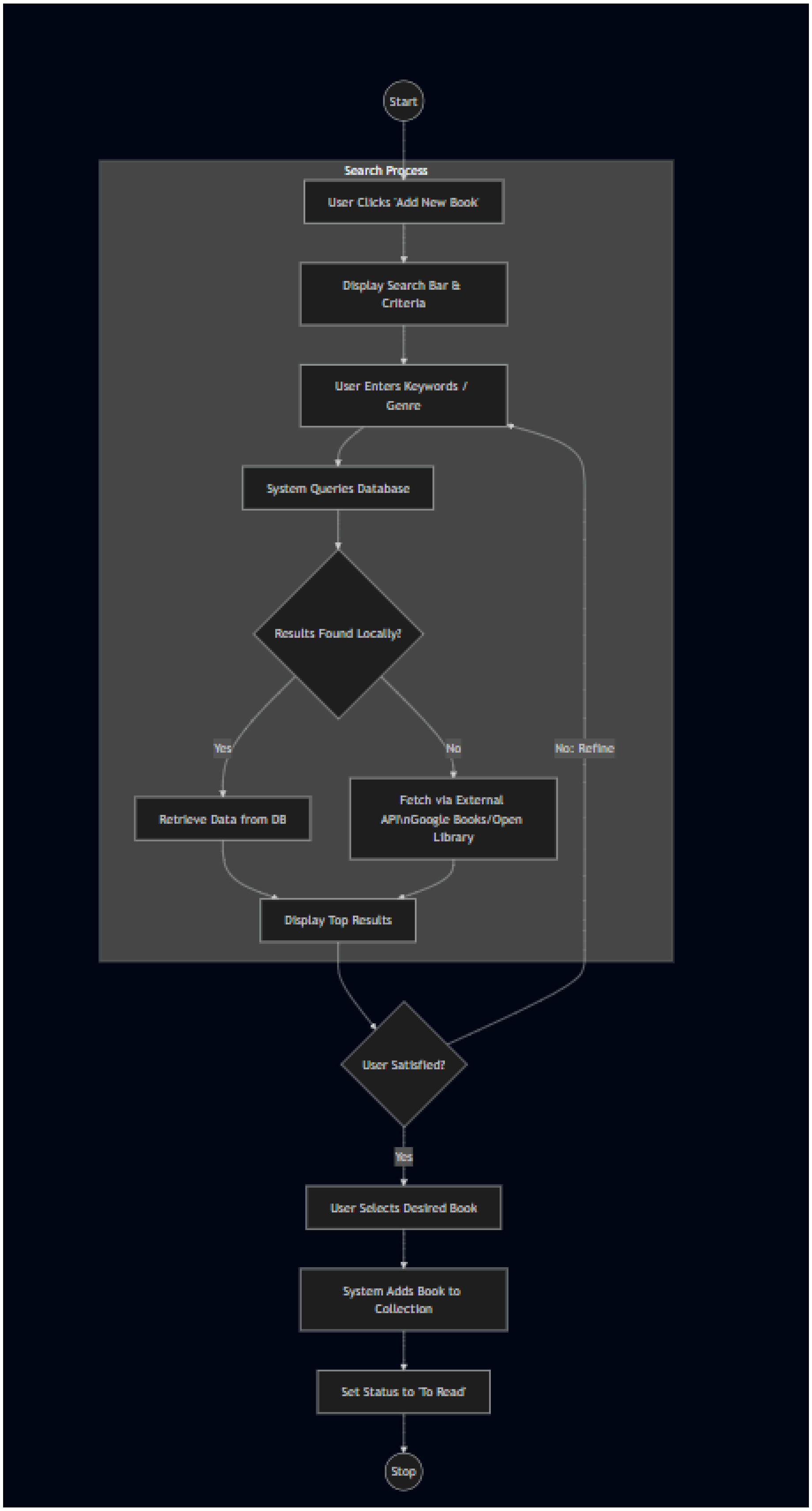


3.4/ Activity Diagrams:

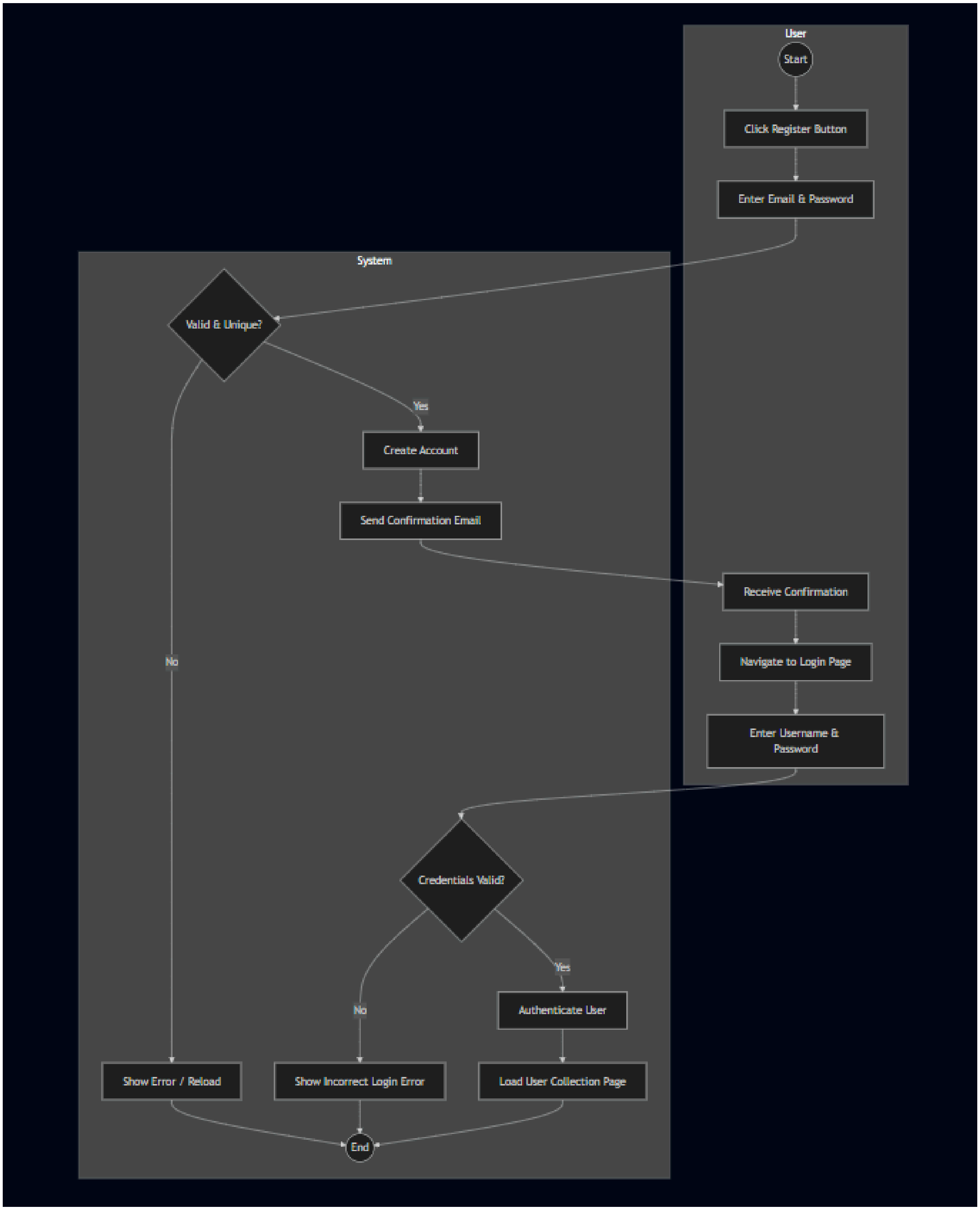
A/ Account Settings:



B/ Book Searching:

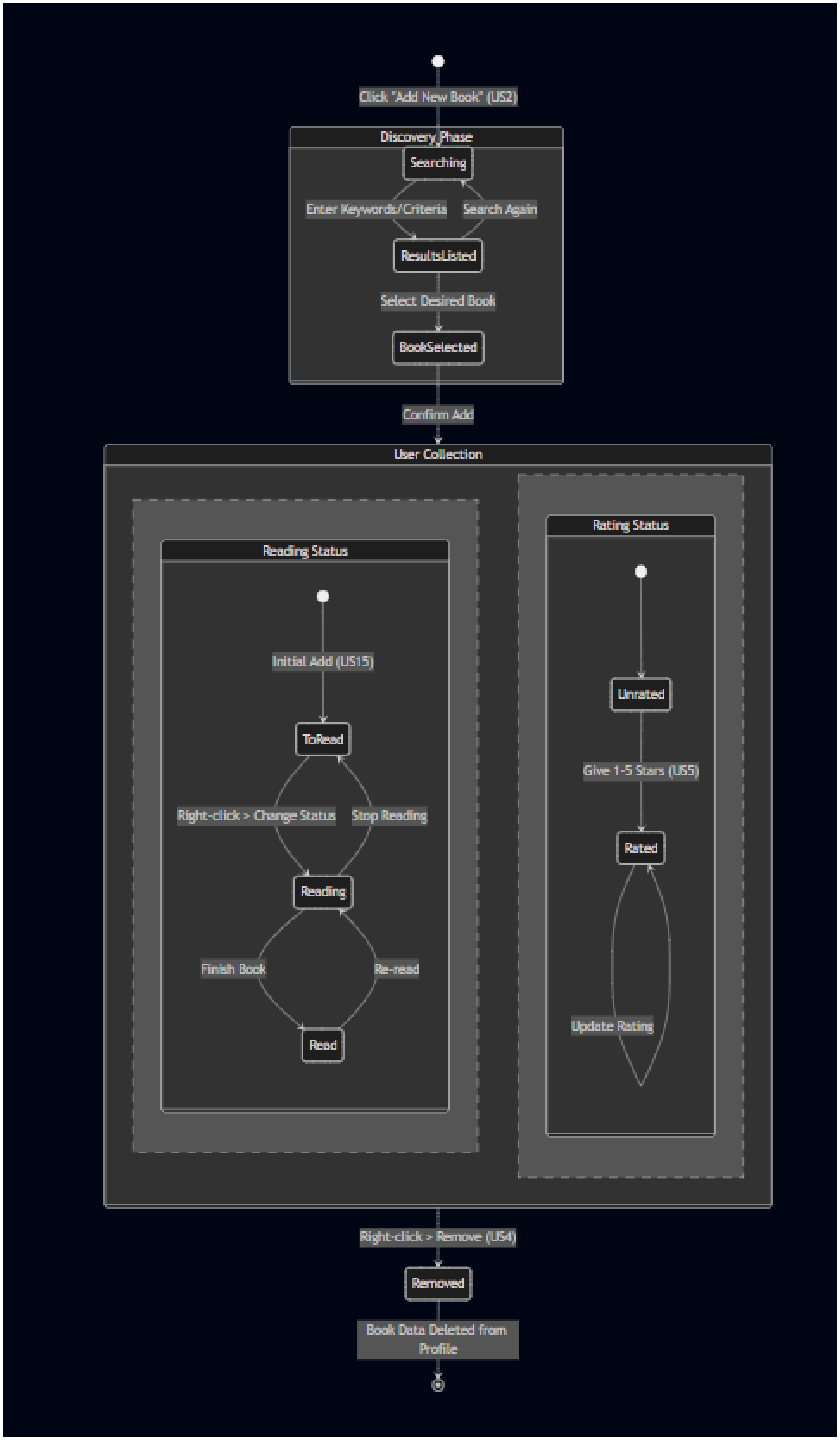


C/ Login and Register:

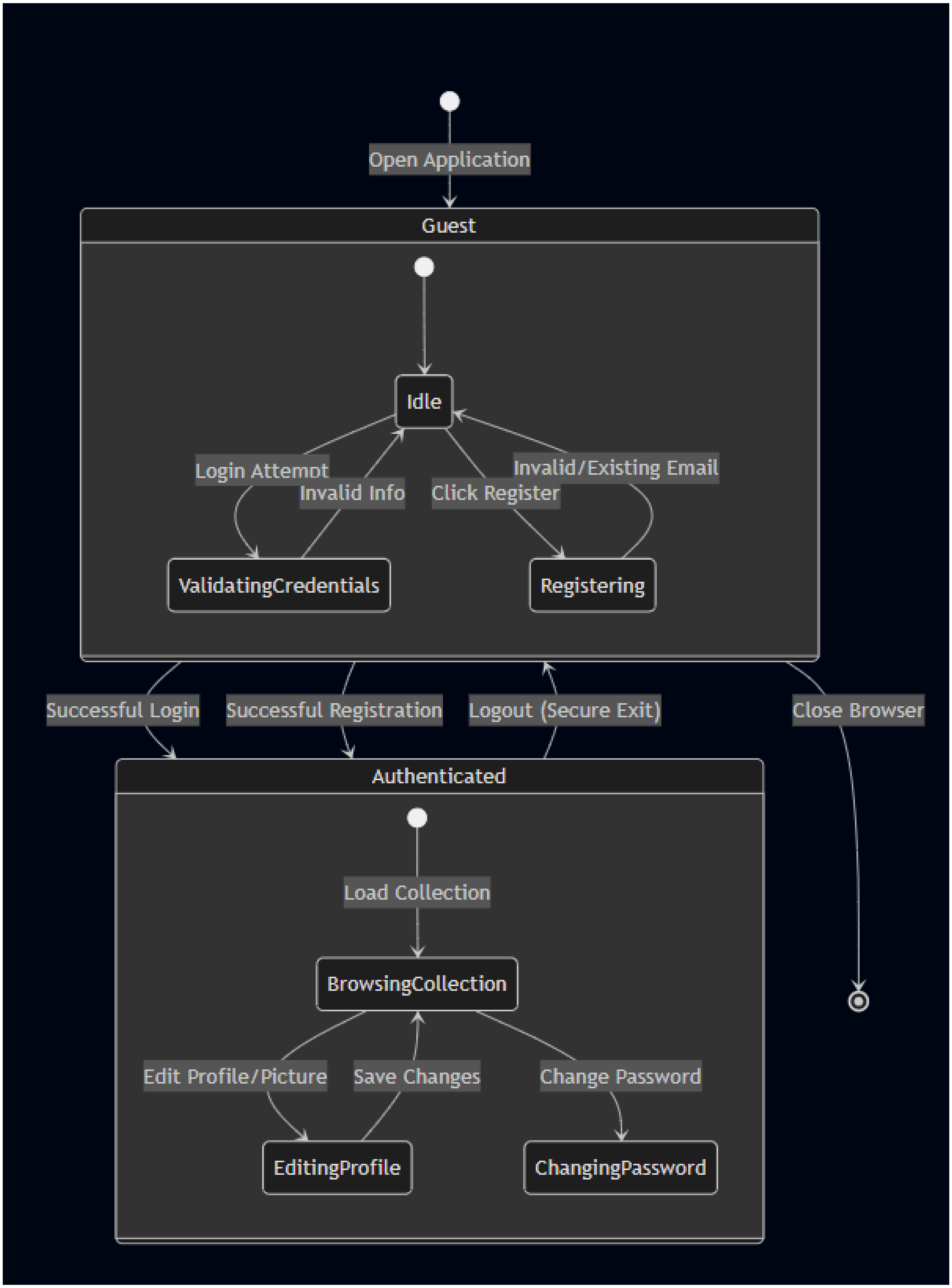


3.5/ State Machine Diagrams:

A/ Book Status and Reading:

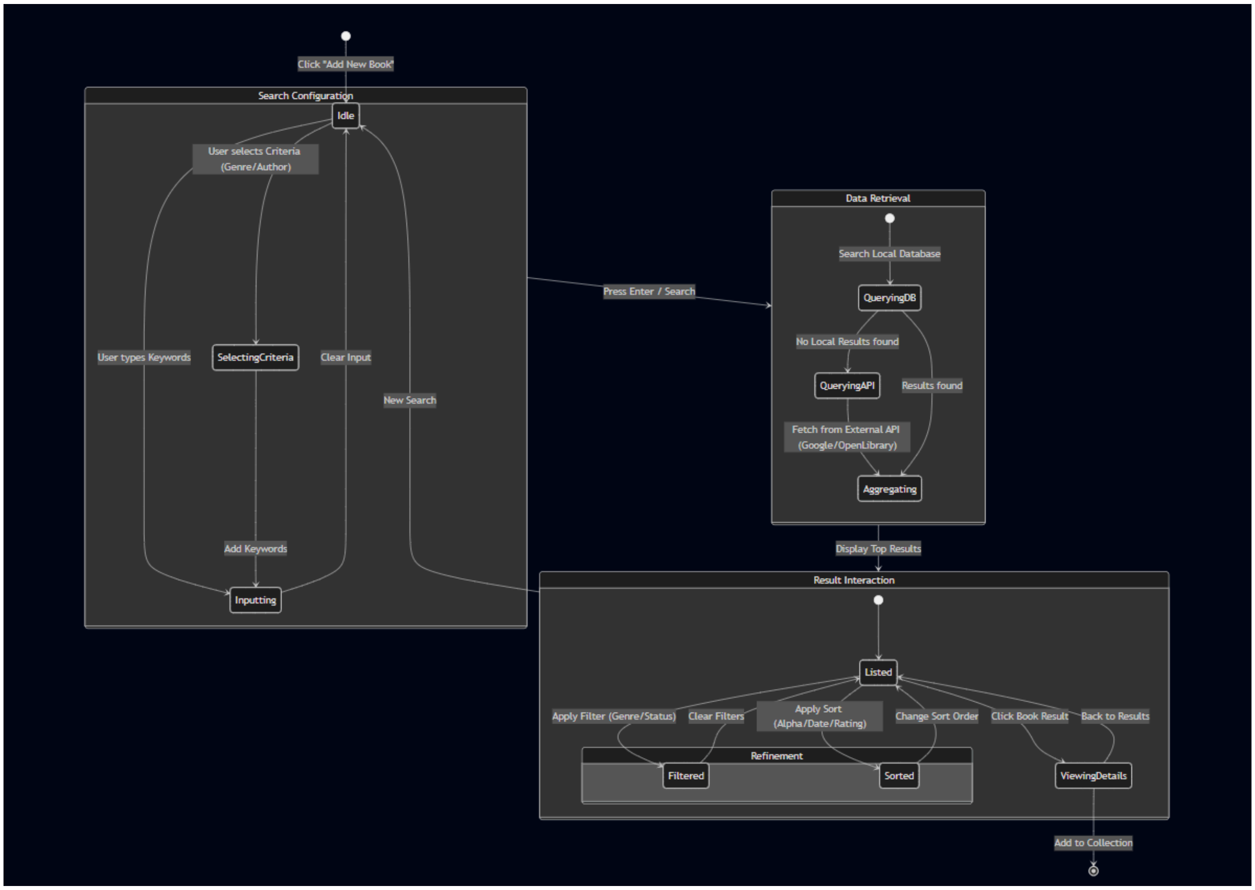


B/ Login:



Open Application
Guest
Idle
Login Attempt
Invalid Info
Click Register
Invalid/Existing Email
ValidatingCredentials
Registering
Successful Login
Successful Registration
Logout (Secure Exit)
Close Browser
Authenticated
Load Collection
BrowsingCollection
Edit Profile/Picture
Save Changes
Change Password
EditingProfile
ChangingPassword

C/ Search:



4. Additional Links:

4.1/ Diagram Folder Github Link:

- https://github.com/memoryVoid1/BookHeaven_Deliverables/tree/main/UML%20Diagrams